II.

Step 1: Create a Deployment

deploy.yml

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: frontend-deploy
spec:
  replicas: 3
  selector:
    matchLabels:
      tier: my-matching-label
  template:
    metadata:
      labels:
        tier: my-matching-label
    spec:
      containers:
      - name: php-redis
        image: gcr.io/google_samples/gb-frontend:v3
```

```
maissa-ubunto@maissa-ubunto-VMware-Virtual-Platform:~$ kubectl create -f deploy.yml
deployment.apps/frontend-deploy created
```

```
maissa-ubunto@maissa-ubunto-VMware-Virtual-Platform:~$ kubectl get deploy
NAME              READY   UP-TO-DATE   AVAILABLE   AGE
frontend-deploy   0/3     3            0           119s
maissa-ubunto@maissa-ubunto-VMware-Virtual-Platform:~$ kubectl get rs
NAME                        DESIRED   CURRENT   READY   AGE
frontend-deploy-6b66b4fcb7  3         3         0       2m14s
newpods                     3         3         3       4d2h
nginx-rs                    2         2         1       3d18h
maissa-ubunto@maissa-ubunto-VMware-Virtual-Platform:~$
```

Step 2: Compare ReplicaSets and Deployments

```
maissa-ubunto@maissa-ubunto-VMware-Virtual-Platform:~$ kubectl get deploy -o yaml > dep.yaml
maissa-ubunto@maissa-ubunto-VMware-Virtual-Platform:~$ kubectl get rs -o yaml > rs.yaml
maissa-ubunto@maissa-ubunto-VMware-Virtual-Platform:~$ diff dep.yaml rs.yaml
4c4
<   kind: Deployment
---
>   kind: ReplicaSet
6a7,8
>       deployment.kubernetes.io/desired-replicas: "3"
>       deployment.kubernetes.io/max-replicas: "4"
10c12,15
<     name: frontend-deploy
---
>     labels:
>       pod-template-hash: 6b66b4fcb7
>       tier: my-matching-label
>     name: frontend-deploy-6b66b4fcb7
12,13c17,25
<     resourceVersion: "15420"
<     uid: 95bb2c88-e409-4ecb-a092-5550ed2bec36
---
>     ownerReferences:
>     - apiVersion: apps/v1
>       blockOwnerDeletion: true
>       controller: true
>       kind: Deployment
>       name: frontend-deploy
>       uid: 95bb2c88-e409-4ecb-a092-5550ed2bec36
>     resourceVersion: "15417"
>     uid: aaf1758f-ba2e-40b5-ae76-33c6e9129e4e
15d26
<     progressDeadlineSeconds: 600
17d27
<     revisionHistoryLimit: 10
```

```
,25d31
    strategy:
      rollingUpdate:
        maxSurge: 25%
        maxUnavailable: 25%
      type: RollingUpdate
a35
          pod-template-hash: 6b66b4fcb7
,56c51
    conditions:
```

Step 3: Scale-up the Number of Replicas

```
maissa-ubunto@maissa-ubunto-VMware-Virtual-Platform:~$ kubectl get rs
NAME                          DESIRED   CURRENT   READY   AGE
frontend-deploy-6b66b4fcb7    3         3         0       8m43s
newpods                       3         3         3       4d2h
nginx-rs                      2         2         1       3d18h
maissa-ubunto@maissa-ubunto-VMware-Virtual-Platform:~$ kubectl scale deploy frontend-deploy --replicas 5
deployment.apps/frontend-deploy scaled
maissa-ubunto@maissa-ubunto-VMware-Virtual-Platform:~$ kubectl get rs
NAME                          DESIRED   CURRENT   READY   AGE
frontend-deploy-6b66b4fcb7    5         5         0       10m
newpods                       3         3         3       4d2h
nginx-rs                      2         2         1       3d18h
maissa-ubunto@maissa-ubunto-VMware-Virtual-Platform:~$
```

Step 4: Change Image

Step 4.1: Change Image of a ReplicaSet

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: frontend-deploy
spec:
  replicas: 3
  selector:
    matchLabels:
      tier: my-matching-label
  template:
    metadata:
      labels:
        tier: my-matching-label
    spec:
      containers:
      - name: redis-container
        image: redis:latest
```

```
maissa-ubunto@maissa-ubunto-VMware-Virtual-Platform:~$ kubectl create -f frontend-minimalistic-rs.yaml
deployment.apps/frontend-deploy created
maissa-ubunto@maissa-ubunto-VMware-Virtual-Platform:~$
```

```
maissa-ubunto@maissa-ubunto-VMware-Virtual-Platform:~$ kubectl describe pods | grep image
  Normal  Pulling    24s   kubelet            Pulling image "redis:latest"
  Normal  Pulling    24s   kubelet            Pulling image "redis:latest"
  Normal  Pulled     23s   kubelet            Successfully pulled image "redis:latest" in 883ms (883ms including wait
ing). Image size: 53257562 bytes.
  Normal  Pulling    24s   kubelet            Pulling image "redis:latest"
  Normal   Pulling    4m48s (x5 over 7m54s)   kubelet            Pulling image "gcr.io/google_samples/gb-frontend:v3"
  Warning  Failed     4m48s (x5 over 7m49s)   kubelet            Failed to pull image "gcr.io/google_samples/gb-front
end:v3": rpc error: code = NotFound desc = failed to pull and unpack image "gcr.io/google_samples/gb-frontend:v3": fa
iled to resolve reference "gcr.io/google_samples/gb-frontend:v3": gcr.io/google_samples/gb-frontend:v3: not found
  Normal   BackOff    2m13s (x21 over 7m48s)  kubelet            Back-off pulling image "gcr.io/google_samples/gb-fro
ntend:v3"
  Normal   Pulling    5m1s (x5 over 7m54s)    kubelet            Pulling image "gcr.io/google_samples/gb-frontend:v3"
  Warning  Failed     5m1s (x5 over 7m50s)    kubelet            Failed to pull image "gcr.io/google_samples/gb-front
end:v3": rpc error: code = NotFound desc = failed to pull and unpack image "gcr.io/google_samples/gb-frontend:v3": fa
iled to resolve reference "gcr.io/google_samples/gb-frontend:v3": gcr.io/google_samples/gb-frontend:v3: not found
  Normal   BackOff    2m26s (x21 over 7m50s)  kubelet            Back-off pulling image "gcr.io/google_samples/gb-fro
ntend:v3"
  Normal   Pulling    4m54s (x5 over 7m54s)   kubelet            Pulling image "gcr.io/google_samples/gb-frontend:v3"
  Warning  Failed     4m54s (x5 over 7m50s)   kubelet            Failed to pull image "gcr.io/google_samples/gb-front
end:v3": rpc error: code = NotFound desc = failed to pull and unpack image "gcr.io/google_samples/gb-frontend:v3": fa
iled to resolve reference "gcr.io/google_samples/gb-frontend:v3": gcr.io/google_samples/gb-frontend:v3: not found
  Normal   BackOff    2m34s (x21 over 7m50s)  kubelet            Back-off pulling image "gcr.io/google_samples/gb-fro
ntend:v3"
  Normal  Pulled          3d19h                   kubelet  Successfully pulled image "busybox" in 1.269s (1.269s inclu
ding waiting). Image size: 2222002 bytes.
  Normal  Pulled          3d19h                   kubelet  Successfully pulled image "busybox" in 1.016s (1.016s inclu
ding waiting). Image size: 2222002 bytes.
  Normal  Pulled          3d18h                   kubelet  Successfully pulled image "busybox" in 1.281s (1.281s inclu
ding waiting). Image size: 2222002 bytes.
```

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: frontend-deploy
spec:
  replicas: 3
  selector:
    matchLabels:
      tier: my-matching-label
  template:
    metadata:
      labels:
        tier: my-matching-label
    spec:
      containers:
      - name: redis-container
        image: redis:5
```

```
maissa-ubunto@maissa-ubunto-VMware-Virtual-Platform:~$ kubectl describe pods | grep image
  Normal  Pulling    12s   kubelet            Pulling image "redis:7"
  Normal  Pulling         13m   kubelet  Pulling image "redis:latest"
  Normal  Pulled          12m   kubelet  Successfully pulled image "redis:latest" in 1m2.722s (1m2.723s including waiting). Image
 size: 53262108 bytes.
  Normal  Pulling         13m   kubelet  Pulling image "redis:latest"
  Normal  Pulled          12m   kubelet  Successfully pulled image "redis:latest" in 56.881s (56.881s including waiting). Image si
 ze: 53262108 bytes.
  Normal  Pulling         13m   kubelet  Pulling image "redis:latest"
  Normal  Pulled          12m   kubelet  Successfully pulled image "redis:latest" in 2.021s (1m3.206s including waiting). Image si
 ze: 53262108 bytes.
  Normal   Pulling         43h (x76 over 2d8h)    kubelet  Pulling image "gcr.io/google_samples/gb-frontend:v3"
  Warning  Failed          43h (x72 over 2d8h)    kubelet  Failed to pull image "gcr.io/google_samples/gb-frontend:v3": rpc error:
 code = NotFound desc = failed to pull and unpack image "gcr.io/google_samples/gb-frontend:v3": failed to resolve reference "gcr.i
o/google_samples/gb-frontend:v3": gcr.io/google_samples/gb-frontend:v3: not found
  Normal   BackOff         43h (x1675 over 2d8h)  kubelet  Back-off pulling image "gcr.io/google_samples/gb-frontend:v3"
  Normal   Pulling         9m25s (x5 over 13m)    kubelet  Pulling image "gcr.io/google_samples/gb-frontend:v3"
```

```
maissa-ubunto@maissa-ubunto-VMware-Virtual-Platform:~$ kubectl get rs
NAME                          DESIRED   CURRENT   READY   AGE
frontend-deploy-599d86d7b6    2         2         1       78s
frontend-deploy-66487599dd    2         2         2       2d8h
frontend-rs                   3         3         0       2d8h
newpods                       3         3         3       6d11h
nginx-rs                      2         2         1       6d3h
maissa-ubunto@maissa-ubunto-VMware-Virtual-Platform:~$
```

VII.Step 5: Rollout, Rollback

```
maissa-ubunto@maissa-ubunto-VMware-Virtual-Platform:~$ kubectl rollout history deployment frontend-deploy
deployment.apps/frontend-deploy
REVISION  CHANGE-CAUSE
1         <none>
2         <none>

maissa-ubunto@maissa-ubunto-VMware-Virtual-Platform:~$
```

a.Step 5.1: Rollout Undo

```
maissa-ubunto@maissa-ubunto-VMware-Virtual-Platform:~$ kubectl rollout undo deployment frontend-deploy
deployment.apps/frontend-deploy rolled back
maissa-ubunto@maissa-ubunto-VMware-Virtual-Platform:~$ kubectl describe pods | grep image
  Normal  Pulling    3m53s  kubelet            Pulling image "redis:7"
  Normal  Pulled     3m6s   kubelet            Successfully pulled image "redis:7" in 46.701s (46.701s including waiting). Image s
ize: 45047408 bytes.
  Normal  Pulling    3s    kubelet            Pulling image "redis:latest"
  Normal  Pulled     1s    kubelet            Successfully pulled image "redis:latest" in 1.823s (1.824s including waiting). Image
 size: 53262108 bytes.
  Normal  Pulling    11s   kubelet            Pulling image "redis:latest"
  Normal  Pulled     8s    kubelet            Successfully pulled image "redis:latest" in 3.613s (3.613s including waiting). Image
 size: 53262108 bytes.
  Normal  Pulling    18s   kubelet            Pulling image "redis:latest"
  Normal  Pulled     15s   kubelet            Successfully pulled image "redis:latest" in 2.167s (2.167s including waiting). Image
 size: 53262108 bytes.
  Normal   Pulling         43h (x76 over 2d8h)    kubelet  Pulling image "gcr.io/google_samples/gb-frontend:v3"
  Warning  Failed          43h (x72 over 2d8h)    kubelet  Failed to pull image "gcr.io/google_samples/gb-frontend:v3": rpc error:
 code = NotFound desc = failed to pull and unpack image "gcr.io/google_samples/gb-frontend:v3": failed to resolve reference "gcr.i
Corbeille samples/gb-frontend:v3": gcr.io/google_samples/gb-frontend:v3: not found
  Normal   BackOff         43h (x1675 over 2d8h)  kubelet  Back-off pulling image "gcr.io/google_samples/gb-frontend:v3"
  Normal   Pulling         14m (x5 over 18m)      kubelet  Pulling image "gcr.io/google_samples/gb-frontend:v3"
  Warning  Failed          14m (x5 over 17m)      kubelet  Failed to pull image "gcr.io/google_samples/gb-frontend:v3": rpc error:
 code = NotFound desc = failed to pull and unpack image "gcr.io/google_samples/gb-frontend:v3": failed to resolve reference "gcr.i
o/google_samples/gb-frontend:v3": gcr.io/google_samples/gb-frontend:v3: not found
```